CMAI 480

Computer Vision

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Introduction to Camera Calibration

Chapter 11

Camera Calibration

- How to estimate the intrinsic and extrinsic parameters of a camera?

REVIEW: ABSOLUTE CONIC

Review: Absolute Conic

- Absolute Conic Ω_∞: the intersection between every sphere and plane at ∞.
- $\vec{x}^T \Omega_\infty \vec{x} = 0$
- Suppose we have a camera matrix $P = KR[I|-\tilde{C}]$
- $\vec{x} = P\vec{X} = KR[I|-\tilde{C}]\vec{X}$
- Let $\vec{X} = \begin{bmatrix} \vec{x_d} \\ 0 \end{bmatrix}$ be a point at infinity on π_∞ (last coordinate is zero).
  - $\therefore \vec{x} = KR\vec{x_d}$, where $\vec{x_d}$ is a 3×1 physical coordinates and KR is a homography H.
  - Note that it does not matter where the camera is but it matters what its direction or orientation is.

Review: Absolute Conic

- Therefore we can write $\vec{x}_{3\times1} = H_{3\times3}(\vec{x}_d)_{3\times1}$
- **Question: How many DoF $(\vec{x}_d)_{3\times1}$ have?**
  - It should be 3 DoF but in fact it has only 2 DoF by insisting that the norm of $\vec{x}_d$ be one i.e. $\|\vec{x_d}\| = 1$.
    - This does not effect our results.
- Now for the absolute conic Ω_∞ the direction $\vec{x}_d$ obey $\vec{x}_d^T \Omega_\infty \vec{x}_d = 0$ or $\vec{x}_d^T I_{3\times3} \vec{x}_d = 0$.

Review: Absolute Conic

- We have $\vec{x}_{3\times1} = H_{3\times3}(\vec{x}_d)_{3\times1}$ a mapping from π_∞(the $\vec{x}_d$ point) to sensor (image) plane ($\vec{x}$).
  - $\vec{x}_{3\times1}$: Homogeneous representation
  - $H_{3\times3}$:Homogeneous representation
  - $(\vec{x}_d)_{3\times1}$: Physical representation
- **Question: How does a conic transform through a homography H?**
  - $C' = H^{-T}CH^{-1}$
- Substituting the absolute conic $\Omega_\infty = I_{3\times3}$ in place of C in the equation $C' = H^{-T}CH^{-1}$, we get $C' = H^{-T}IH^{-1}$ as the image of the absolute conic in the sensor or image plane $\Rightarrow$

Review: Absolute Conic

- But, also we already know that $H = KR$
- Therefore, $C' = \omega = (KR)^{-T} I (KR)^{-1} = (R^{-1}K^{-1})^T (R^{-1}K^{-1}) = K^{-T}R^{-T}R^{-1}K^{-1} = K^{-T}K^{-1}$
  - Note that $R^{-T}R^{-1} = I$ because R is a rotation matrix (orthogonal non-singular and is of full rank matrix).
  - **Note that $\omega = K^{-T}K^{-1}$ is the very real image of some something very unreal.**
- Recall that $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow K^{-1} = \begin{bmatrix} 1/\alpha_x & 0 & -x_0/\alpha_x \\ 0 & 1/\alpha_y & -y_0/\alpha_y \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow K^{-T} = \begin{bmatrix} 1/\alpha_x & 0 & 0 \\ 0 & 1/\alpha_y & 0 \\ -x_0/\alpha_x & -y_0/\alpha_y & 1 \end{bmatrix}$
- $\omega = K^{-T}K^{-1} = \begin{bmatrix} 1/\alpha_x & 0 & 0 \\ 0 & 1/\alpha_y & 0 \\ -x_0/\alpha_x & -y_0/\alpha_y & 1 \end{bmatrix} \begin{bmatrix} 1/\alpha_x & 0 & -x_0/\alpha_x \\ 0 & 1/\alpha_y & -y_0/\alpha_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\alpha_x^2} & 0 & \frac{-x_0}{\alpha_x^2} \\ 0 & \frac{1}{\alpha_y^2} & \frac{-y_0}{\alpha_y^2} \\ \frac{-x_0}{\alpha_x^2} & \frac{-y_0}{\alpha_y^2} & \frac{x_0^2}{\alpha_x^2} + \frac{y_0^2}{\alpha_y^2} + 1 \end{bmatrix}$

Review: Absolute Conic

- But ω is a conic that has 5 DoF given by

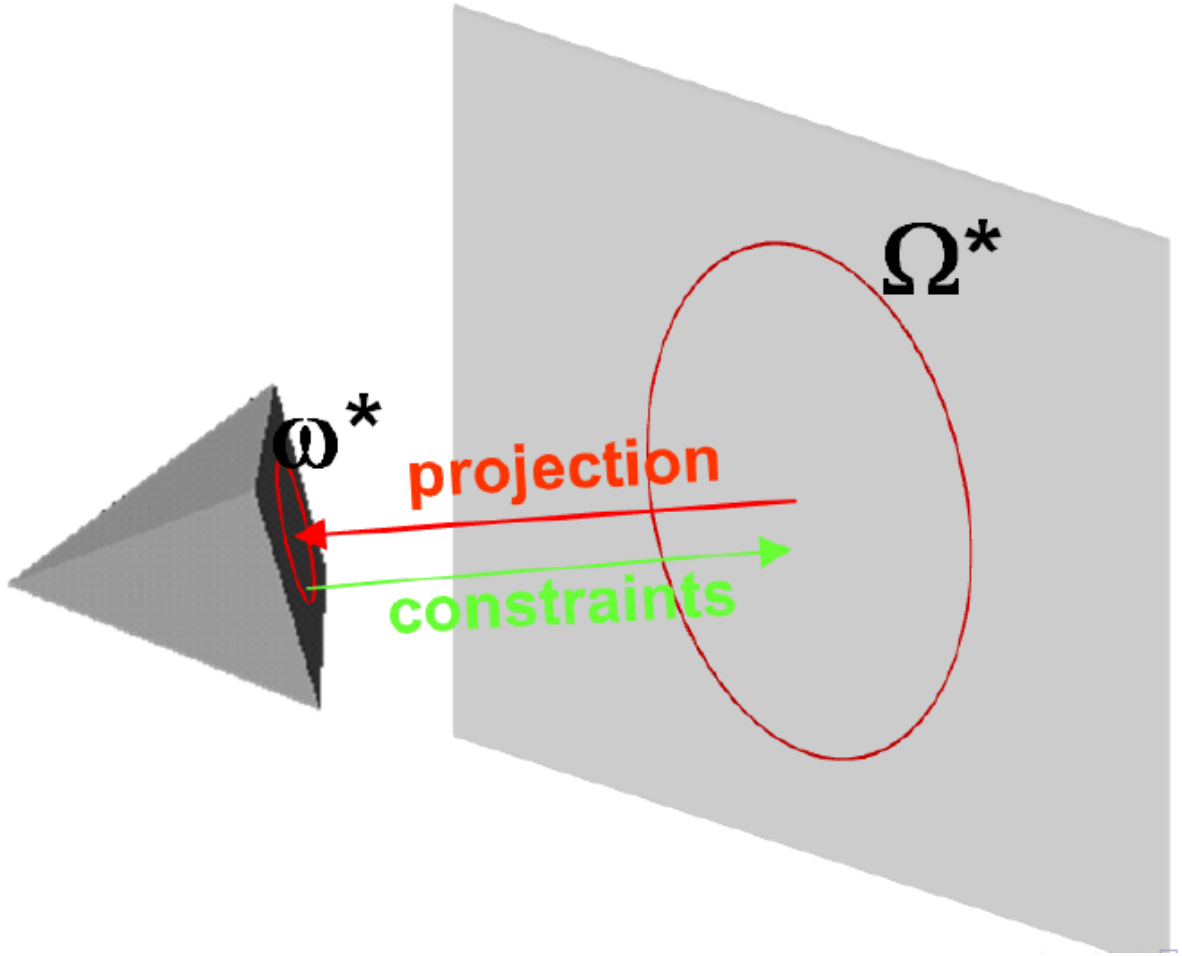
$$\begin{bmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{bmatrix} == \begin{bmatrix} \frac{1}{\alpha_x^2} & 0 & \frac{-x_0}{\alpha_x^2} \\ 0 & \frac{1}{\alpha_y^2} & \frac{-y_0}{\alpha_y^2} \\ \frac{-x_0}{\alpha_x^2} & \frac{-y_0}{\alpha_y^2} & \frac{x_0^2}{\alpha_x^2} + \frac{y_0^2}{\alpha_y^2} + 1 \end{bmatrix}$$

  - $ax^2 + bxy + cy^2 + dx + ey + f = 0$
- Thus:
  - $a = \frac{1}{\alpha_x^2}$, $b = 0$, $d = \frac{-2x_0}{\alpha_x^2}$, $c = \frac{1}{\alpha_y^2}$, $e = \frac{-2y_0}{\alpha_y^2}$, $f = \frac{x_0^2}{\alpha_x^2} + \frac{y_0^2}{\alpha_y^2} + 1$
- Conclusion if you would be able to **find ω** then you can find the intrinsic parameters of K using the above equations **or you can use Cholsky Decomposition**.
- Therefore, our main goal now is how to find ω (the image of the absolute conic Ω_∞)
  - Need at least five points.

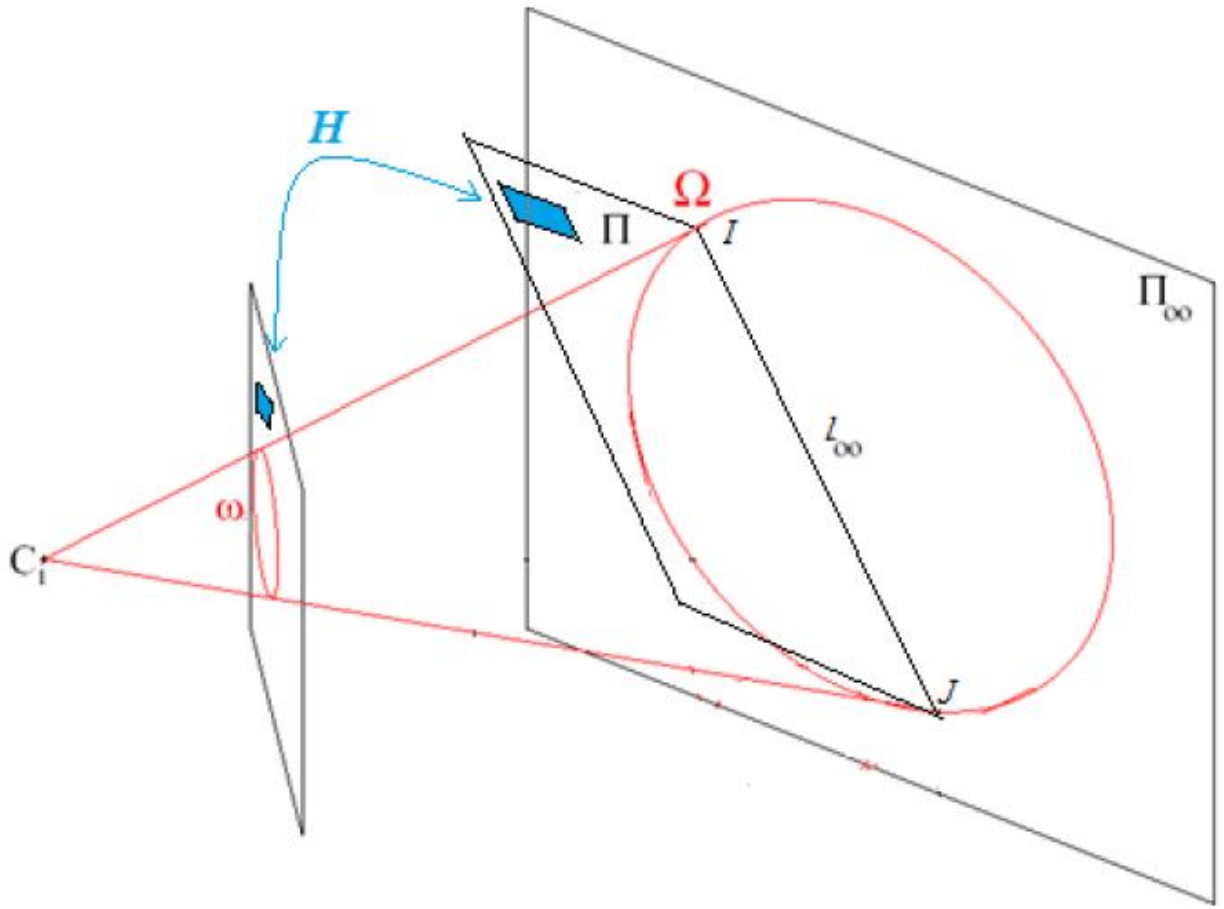
The image of the Absolute Conic Ω_∞: ω

- **Question: How do we find the image points in ω and therefore ω?**
  - We probe the absolute conic Ω_∞ with a plane $\vec{\Pi}$
  - We use the fact that any $\vec{\Pi}$ will intersect Ω_∞ at exactly two points — the circular points of $\vec{\Pi}_\infty$
    - Recall: the circular points are the intersection of every circle (conic) with l_∞ in 2D.
  - Let H be the homography between the probe plane $\vec{\pi}$ and the camera sensor plane that can be computed easily using at least 4 non-collinear correspondences by showing a pattern to the camera.
  - Now we will use the fact that the image of the absolute conic Ω_∞ is ω on the sensor plane and since the circular points are originally on the absolute conic Ω_∞, the image of these circular points will be on ω. Using this fact and the known homography H, we now will show how to find ω.

The image of the Absolute Conic Ω_∞: ω



The image of the Absolute Conic Ω_∞: ω



The image of the Absolute Conic Ω_∞: ω

- Using H : the homography between the probe plane $\vec{\pi}$ and the camera sensor plane that can be computed using at least 4 non-collinear correspondences, we can image the circular points that are given by:
  - $I = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, J = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$
- Image of the first circular point (imaginary):

$$\vec{x}_1 = HI = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \vec{h}_1 + i\vec{h}_2$$

- Image of the second circular point (imaginary):

$$\vec{x}_2 = HJ = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} = \vec{h}_1 - i\vec{h}_2$$

The image of the Absolute Conic Ω_∞: ω

- But we know that both of $\vec{x}_1$ and $\vec{x}_2$ must fall on the conic $\omega = K^{-T}K^{-1}$ and thus must satisfy the equation of the conic:
  - $\vec{x}_1^T \omega \vec{x}_1 = 0 \Rightarrow \left(\vec{h}_1 + i\vec{h}_2\right)^T \omega \left(\vec{h}_1 + i\vec{h}_2\right) = 0$, and
  - $\vec{x}_2^T \omega \vec{x}_2 = 0 \Rightarrow \left(\vec{h}_1 - i\vec{h}_2\right)^T \omega \left(\vec{h}_1 - i\vec{h}_2\right) = 0$

The image of the Absolute Conic Ω_∞: ω

- Therefore, we have:
  - $\vec{h}_1^T \omega \vec{h}_1 - \vec{h}_2^T \omega \vec{h}_2 + i\left(\vec{h}_1^T \omega \vec{h}_2 + \vec{h}_2^T \omega \vec{h}_1\right) = 0$, and
  - $\vec{h}_1^T \omega \vec{h}_1 - \vec{h}_2^T \omega \vec{h}_2 - i\left(\vec{h}_1^T \omega \vec{h}_2 + \vec{h}_2^T \omega \vec{h}_1\right) = 0$
- Since ω is symmetry we must have:
  - $\vec{h}_1^T \omega \vec{h}_2 = \vec{h}_2^T \omega \vec{h}_1 = 0$ ...(1)
- As a result we also should have
  - $\vec{h}_1^T \omega \vec{h}_1 = \vec{h}_2^T \omega \vec{h}_2$ ...(2)
- Recall that $\vec{h}_1$ and $\vec{h}_2$ are part of the homography obtained by showing one pattern to the camera.
- Thus, we have two equations but 5 unknowns of ω .
- As such, we need at least three probing planes (or three different computed homographies) to estimate ω.

The image of the Absolute Conic Ω_∞: ω

- Methodology:
  - Setup your camera
  - Move at least three planes with respect to the camera
    - `https://docs.opencv.org/3.4/da/d0d/tutorial_camera_calibration_pattern.html`
  - That's all !!!!
- **Question: Could we just show one probe plane to the camera and calibrate it?**
  - Ans: Yes, provided your probe plane is rich with patterns.
    - It is a Research Area.

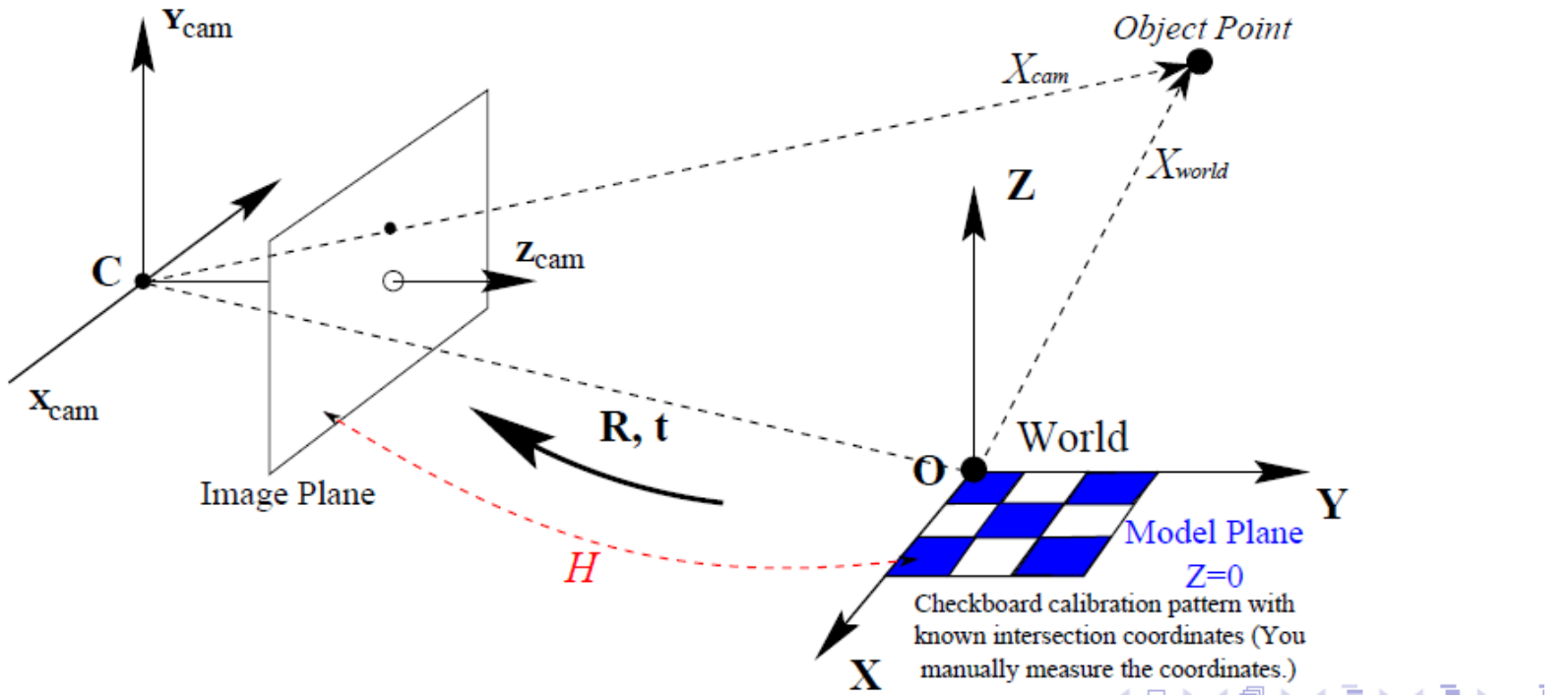
Camera Calibration: Zhang's Algorithm

Camera Calibration: Zhang's Algorithm

- The **camera calibration algorithm** developed originally by **Zhengyyou Zhang** (currently leading both the Robotic X Lab and AI Lab at Tencent after 20 years of work at Microsoft Research) on the basis of the ideas discussed in the previous lecture (or in the previous slides) **is the world's most popular such algorithm today.**

Zhengyou Zhang's Calibration Method

- Refer to "**A Flexible New Technique for Camera Calibration**" at `http://research.microsoft.com/en-us/um/people/zhang/calib/`
  - A technique to figure out both of the intrinsic (K) and extrinsic parameters (R, t) of the camera



Zhengyou Zhang's Calibration Method

- Recall camera imaging:
  - $\vec{x} = KR\left[I|-\vec{C}\right]\vec{X}_{world}$ or $\vec{x} = K\left[R|\vec{t}\right]\vec{X}_{world}$
  - $\vec{x} = K\left[I|\vec{0}\right]\vec{X}_{cam}$
- Note that $\vec{X}_{cam} = R\vec{X}_{world} + \vec{t}$, where $\vec{t} = -R\vec{C}$
- Therefore, In a matrix form we can write $\vec{x} = K\left[I|\vec{0}\right]\begin{bmatrix} R & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}\vec{X}_{world}$

Zhengyou Zhang's Calibration Method

- Zhang's uses the camera equation that we showed in the previous slide $\vec{x} = K\left[R|\vec{t}\right]\vec{X}_{world}$ in his calibration method:
  - $\vec{x} = K\left[\begin{array}{cccc}\vec{r}_1 & \vec{r}_2 & \vec{r}_3 & \vec{t}\end{array}\right]\vec{X}_{world}$
  - **Recall that our model (check-board calibration) pattern is in $X-Y$ plane i.e. $Z=0$ of the world coordinate system**
  - Therefore, all of the world points $\vec{X}_{world}$ located on the model pattern will have the following coordinates $\begin{pmatrix} X \\ Y \\ 0 \\ W \end{pmatrix}$
  - As such, we will have

$$\vec{x} = K\left[\begin{array}{cccc}\vec{r}_1 & \vec{r}_2 & \vec{r}_3 & \vec{t}\end{array}\right]\begin{pmatrix} X \\ Y \\ 0 \\ W \end{pmatrix} = K\left[\begin{array}{ccc}\vec{r}_1 & \vec{r}_2 & \vec{t}\end{array}\right]\begin{pmatrix} X \\ Y \\ W \end{pmatrix} = H\begin{pmatrix} X \\ Y \\ W \end{pmatrix}, \text{ where}$$

$$\boldsymbol{H = K\left[\begin{array}{ccc}\vec{r}_1 & \vec{r}_2 & \vec{t}\end{array}\right]}$$

Zhengyou Zhang's Calibration Method

- $H = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{t} \end{bmatrix}$ in the previous slide represents the homography between the image points in the sensor (image) plane and the world points in the model (calibration) plane which can be computed easily by establishing point correspondences between the two planes (image and model planes).
  - So we have now $\vec{x}_{image} = H\vec{X}_{model}$
- If we let the homography H between the image points in the sensor (image) plane and the world points in the model (calibration) plane be given by $H = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix}_{3\times 3}$ then we will have:
  - $\begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix} = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{t} \end{bmatrix}$
  - Therefore,
    - $\vec{r}_1 = K^{-1}\vec{h}_1$
    - $\vec{r}_2 = K^{-1}\vec{h}_2$
    - $\vec{t} = K^{-1}\vec{h}_3$

Zhengyou Zhang's Calibration Method

- So if $\boldsymbol{H} = \left\lfloor \vec{h}_1 \quad \vec{h}_2 \quad \vec{h}_3 \right\rfloor = \boldsymbol{K} \left[\vec{r}_1 \quad \vec{r}_2 \quad \vec{t} \right]$ then we will have:
  - $\vec{r}_1 = K^{-1}\vec{h}_1$, $\vec{r}_2 = K^{-1}\vec{h}_2$, and $\vec{t} = K^{-1}\vec{h}_3$
- However, note that $R = \left[\vec{r}_1 \quad \vec{r}_2 \quad \vec{r}_3 \right]$
  - **Additionally, note that R and $\vec{t}$ defines the extrinsic parameters of the camera.**
- Moreover, Note that R :
  - is an orthonormal matrix or an orthogonal matrix whose norm is equal to one, and
  - all of its column vectors are orthogonal to each others.
- Consequently, $\vec{r}_1 \perp \vec{r}_2$ and $\vec{r}_1^T . \vec{r}_2 = 0$ and $\vec{r}_3$ is orthogonal to both $\vec{r}_1$and $\vec{r}_2$.
  - Therefore, we can get $\vec{r}_3$as the cross product of both $\vec{r}_1$ and $\vec{r}_2$: $\vec{r}_3 = \vec{r}_1 \times \vec{r}_2$

Zhengyou Zhang's Calibration Method

- Recall that $\vec{r}_1 = K^{-1}\vec{h}_1, \quad \vec{r}_2 = K^{-1}\vec{h}_2, \quad \vec{r}_3 = \vec{r}_1 \times \vec{r}_2, \quad \vec{t} = K^{-1}\vec{h}_3$
- Since $\vec{r}_1^T.\vec{r}_2 = 0$, we will have $\left(K^{-1}\vec{h}_1\right)^T \left(K^{-1}\vec{h}_2\right) = 0$
  - $\therefore \vec{h}_1^T K^{-T} K^{-1}\vec{h}_2 = 0$ or $\vec{h}_1^T \omega \vec{h}_2 = 0$ **...(1)**
- Similarly, because $\vec{r}_1^T.\vec{r}_1 = \vec{r}_2^T.\vec{r}_2$ (R is an orthonormal matrix) we can get:
  - $\vec{h}_1^T \omega \vec{h}_1 - \vec{h}_2^T \omega \vec{h}_2 = 0$ ...(2)
- From the above two equations we can estimate the image of absolute conic ω by using at least 3 probing planes (3 homographies) subject to the condition that your probing planes stay in the $x-y$ plane in the world coordinate system.
  - 6 unknowns but 5 DoF of ω : can be estimated by showing three model planes in the x-y plane where each probing plane can provide us with two equations.
- **By knowing ω, we thus can compute K as we have demonstrated before.**

Zhengyou Zhang's Calibration Method

- **Extrinsic parameters:**
  - After computing K, you only need one distinguished view for which you can compute $R = \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{r}_3 \end{bmatrix}$ and $\vec{t}$ using the following equations:
    - $\vec{r}_1 = K^{-1}\vec{h}_1$
    - $\vec{r}_2 = K^{-1}\vec{h}_2$
    - $\vec{r}_3 = \vec{r}_1 \times \vec{r}_2$,
    - $\vec{t} = K^{-1}\vec{h}_3$

Matlab: Camera Calibration

- Please review the Matlab Camera Calibration Documentation at:
  - https://www.mathworks.com/help/vision/camera-calibration.html

Questions?